

D902

Secondary Mathematics Curriculum

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

An eighth-grade teacher whose school follows a rotating A/B schedule sees students struggling to interpret slope as a rate. The school's mascot is a hawk. Which earlier idea in the learning progression is most directly foundational?

- A. Place value in whole numbers.
- B. Unit rates and proportional relationships.
- C. Quadratic functions.
- D. The Pythagorean theorem.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A textbook chapter with a garden theme asks, "Find the area of a 6×4 rectangle." The book is in its fourth edition. Which revision best raises the problem's demand?

- A. "Rewrite it as a garden bed to match the chapter theme."
- B. "Find the area of a 12×8 rectangle."
- C. "Find all whole-number rectangles with area 24. Which has the smallest perimeter? Explain."
- D. "Find the areas of ten different rectangles."

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

Midway through a unit, a teacher who writes her own quizzes gives a short ungraded check and uses the results to form small groups for reteaching the next day. The quiz has six items. What type of assessment is this?

- A. Summative.
- B. Norm-referenced.
- C. Diagnostic pre-assessment.
- D. Formative.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A cart in the physics lab shared with the science department moves with velocity $v(t) = 2t$ meters per second from $t = 0$ to $t = 5$. The cart has a mass of 1.2 kg. How far does it travel?

- A. 25 meters
- B. 2 meters
- C. 10 meters
- D. 50 meters

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

Water flows into a tank at $r(t)$ liters per minute, and $\int_0^{10} r(t) dt = 120$. The tank held 30 liters at $t = 0$. How much does it hold at $t = 10$?

- A. 90 liters
- B. 120 liters
- C. 12 liters
- D. 150 liters

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

For a warm-up written on the back whiteboard, a teacher asks students to simplify i^{23} . The class had just returned from a field trip. What is the value?

- A. -1
- B. $-i$
- C. $23i$
- D. i

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

What is $(2 + 3i)(2 - 3i)$?

- A. 13
- B. -5
- C. $-5 + 12i$
- D. 4

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

On a map of a park that covers 40 acres, a bench is at (1, 2) and a fountain is at (7, 10), with units in meters. The map was updated last spring. What is the straight-line distance between them?

- A. 100 meters
- B. About 14.4 meters
- C. 10 meters
- D. 14 meters

Target word _____	The one fact that decides _____	Answer ____	☐ sure ☐ unsure ☐ guess
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9. [TRANSFER — reversed stem / different variable]

The distance between (2, k) and (5, 1) is 5, and $k > 1$. What is k?

- A. -3
- B. 5
- C. 6
- D. 4

Target word _____	The one fact that decides _____	Answer ____	☐ sure ☐ unsure ☐ guess
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10.

A class fits $\hat{y} = 2x + 10$ to data on study hours and quiz scores, with $r = 0.87$. One student's point is (5, 25). The data were collected in October. What is the residual for that point?

- A. 25
- B. 20
- C. -5
- D. 5

Target word _____	The one fact that decides _____	Answer ____	☐ sure ☐ unsure ☐ guess
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11. [TRANSFER — reversed stem / different variable]

For the line $\hat{y} = 1.5x + 2$, a point at $x = 4$ has a residual of -3. What is the observed y?

- A. 5
- B. 8
- C. -3
- D. 11

Target word _____	The one fact that decides _____	Answer ____	☐ sure ☐ unsure ☐ guess
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12.

A teacher who enjoys hiking writes an item to assess solving linear equations. It is a 200-word story about a camping trip with difficult vocabulary, ending in $3x + 7 = 22$. The item is worth 4 points. What is the main design problem?

- A. None; real-world contexts always improve items.
- B. It needs more equations to be reliable.
- C. The reading load may keep students from showing their algebra skill.
- D. It should be multiple choice instead.

Target word _____	The one fact that decides _____	Answer _____	☐ sure ☐ unsure ☐ guess
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STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

An eighth-grade teacher ~~whose school follows a rotating A/B schedule~~ sees students struggling to interpret slope as a rate. ~~The school's mascot is a hawk.~~ Which earlier idea in the learning progression is most directly foundational?

Target word: prerequisite idea **Decides:** Slope as a rate grows from unit rates and proportional relationships.

A	Adjacent category	Too far back in the progression to be the direct link.
B	Correct	Slope generalizes the constant of proportionality.
C	Inverted operation	A later idea, not a foundation.
D	Confusion set	Same grade, but not what slope is built on.

2. Answer C Concept 2

A textbook chapter ~~with a garden theme~~ asks, "Find the area of a 6×4 rectangle." ~~The book is in its fourth edition.~~ Which revision best raises the problem's demand?

Target word: opening a problem **Decides:** Multiple answers plus a choice and justification.

A	Camouflage	A new setting leaves the same closed task.
B	Wrong stem number	Bigger numbers, same single procedure.
C	Correct	Many answers, a comparison, and a justification.
D	Adjacent category	More practice, not more thinking.

3. Answer D Concept 3

Midway through a unit, a teacher ~~who writes her own quizzes~~ gives a short ungraded check and uses the results to form small groups for reteaching the next day. ~~The quiz has six items.~~ What type of assessment is this?

Target word: assessment purpose **Decides:** Results are used during learning to adjust instruction.

A	Inverted operation	Summative measures learning at the end; this is mid-unit and ungraded.
B	Confusion set	Norm-referenced compares students to a norm group.
C	Adjacent category	A pre-assessment happens before instruction begins.
D	Correct	The data shape the next steps of instruction.

4. Answer A Concept 4

A cart in the physics lab ~~shared with the science department~~ moves with velocity $v(t) = 2t$ meters per second from $t = 0$ to $t = 5$. ~~The cart has a mass of 1.2 kg.~~ How far does it travel?

Target word: displacement from velocity **Decides:** Area under $v(t) = 2t$ from 0 to 5.

A	Correct	$\int_0^5 2t \, dt = t^2$ from 0 to 5 = 25 (a triangle $\frac{1}{2} \times 5 \times 10$).
B	Inverted operation	Differentiates (acceleration) instead of integrating.
C	Confusion set	$v(5) = 10$ is the final velocity.
D	Dropped element	Uses the 5×10 rectangle and drops the $\frac{1}{2}$ for a triangle.

5. Answer D Concept 4 · transfer

Water flows into a tank at $r(t)$ liters per minute, and $\int_0^{10} r(t) dt = 120$. The tank held 30 liters at $t = 0$. How much does it hold at $t = 10$?

Target word: final amount from accumulation **Decides:** Final = initial + net change.

A	Inverted operation	Subtracts the inflow.
B	Dropped element	The net change, without the starting amount.
C	Adjacent category	120/10 is the average inflow rate per minute.
D	Correct	$30 + 120 = 150$.

6. Answer B Concept 5

For a warm-up ~~written on the back whiteboard~~, a teacher asks students to simplify i^{23} . ~~The class had just returned from a field trip.~~ What is the value?

Target word: power of i **Decides:** $23 = 4 \times 5 + 3$, so $i^{23} = i^3$.

A	Wrong stem number	That is i^2 , from a remainder of 2 instead of 3.
B	Correct	$i^3 = i^2 \cdot i = -i$.
C	Inverted operation	Multiplies by 23 instead of raising to the 23rd power.
D	Confusion set	Assumes every odd power of i equals i .

7. Answer A Concept 5 · transfer

What is $(2 + 3i)(2 - 3i)$?

Target word: product of conjugates **Decides:** $a^2 + b^2$ with $i^2 = -1$.

A	Correct	$4 - 9i^2 = 4 + 9 = 13$.
B	Inverted operation	Treats i^2 as $+1$.
C	Confusion set	That is $(2 + 3i)^2$, not the conjugate product.
D	Dropped element	Drops the i^2 term.

8. Answer C Concept 6

On a map of a park ~~that covers 40 acres~~, a bench is at $(1, 2)$ and a fountain is at $(7, 10)$, with units in meters. ~~The map was updated last spring.~~ What is the straight-line distance between them?

Target word: distance between points **Decides:** $\Delta x = 6$, $\Delta y = 8$.

A	Dropped element	Forgets the square root.
B	Inverted operation	Adds coordinates (8 and 12) instead of subtracting them.
C	Correct	$\sqrt{(36 + 64)} = \sqrt{100} = 10$.
D	Adjacent category	$6 + 8$ is the walking distance along the grid, not the straight line.

9. Answer B Concept 6 · transfer

The distance between $(2, k)$ and $(5, 1)$ is 5, and $k > 1$. What is k ?

Target word: solve for a coordinate **Decides:** $3^2 + (k - 1)^2 = 25$, so $k - 1 = \pm 4$.

A	Dropped element	The other root, which breaks the condition $k > 1$.
B	Correct	$k - 1 = 4$ (since $k > 1$), so $k = 5$.
C	Wrong stem number	Uses the full distance 5 as the vertical leg.
D	Confusion set	Reports $k - 1$ instead of k .

10. Answer D Concept 7

A class fits $\hat{y} = 2x + 10$ to data on study hours and quiz scores, with $r = 0.87$. One student's point is $(5, 25)$. The data were collected in October. What is the residual for that point?

Target word: residual **Decides:** $\hat{y} = 2(5) + 10 = 20$.

A	Dropped element	The observed value, with nothing subtracted.
B	Confusion set	That is the predicted value.
C	Inverted operation	Predicted minus observed reverses the order.
D	Correct	$25 - 20 = 5$; the point lies above the line.

11. Answer A Concept 7 · transfer

For the line $\hat{y} = 1.5x + 2$, a point at $x = 4$ has a residual of -3 . What is the observed y ?

Target word: solve for observed y **Decides:** $\hat{y} = 8$ and $y = \hat{y} + \text{residual}$.

A	Correct	$8 + (-3) = 5$.
B	Dropped element	The predicted value, without the residual.
C	Wrong stem number	The residual itself.
D	Inverted operation	Subtracts the residual from $\hat{y}$.

12. Answer C Concept 8

A teacher who enjoys hiking writes an item to assess solving linear equations. It is a 200-word story about a camping trip with difficult vocabulary, ending in $3x + 7 = 22$. The item is worth 4 points. What is the main design problem?

Target word: item validity **Decides:** Reading load is not part of the skill being assessed.

A	Absolute language	Context can add barriers that are not the target skill.
B	Adjacent category	Reliability is a separate issue from what the item measures.
C	Correct	Scores could reflect reading, not equation solving.
D	Confusion set	Changing the format leaves the reading barrier in place.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	7	
Wrong stem number	A figure copied from the prompt that is not the answer.	4	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	1	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Learning progressions and vertical coherence	Standards build across grades (unit rates → proportional relationships → linear functions); gaps are traced back to the idea directly beneath the current one.	Reteaching something unrelated or far earlier, or looking ahead instead of back.	Students struggle with a grade-level idea; “what earlier idea is foundational?”
2	Open vs. closed problems	Opening a problem (many answers, choices to make, justification) raises its demand; bigger numbers, more copies, or a new setting do not.	Thinking harder numbers or a story context makes a problem richer.	“Which revision raises the demand of this problem?”
3	Formative vs. summative assessment	Assessment is formative when its results are used during learning to adjust instruction; summative when it certifies learning at the end.	Classifying an assessment by its format instead of its purpose.	An assessment described by when and why it is given.
4	The integral as accumulation	The integral of a rate over an interval gives the net change in the quantity; the area under a velocity graph is displacement.	Reading the final rate as the total, or differentiating when the question needs accumulation.	A rate function and “how far” or “how much in total.”
5	Complex numbers and powers of i	$i^2 = -1$, so powers of i cycle i , -1 , $-i$, 1 ; $(a + bi)(a - bi) = a^2 + b^2$.	Treating i^2 as $+1$, or guessing odd powers of i always equal i .	A power of i or a product of conjugates.
6	The distance formula is the Pythagorean theorem	The distance between two points is $\sqrt{(\Delta x)^2 + (\Delta y)^2}$, the hypotenuse of a right triangle.	Adding the legs, or forgetting the square root.	Two coordinate points and “how far apart?”
7	Residuals	Residual = observed y – predicted $\hat{y}$; a positive residual means the point lies above the line.	Subtracting in the wrong order, or reporting the predicted value.	A fitted line and one data point.
8	Items should measure the intended skill	A well-designed problem removes barriers that are not part of the skill being assessed, such as heavy reading load in a math item.	Assuming any real-world context improves an item.	A math item with a long or complex text wrapper.

Course scope (public description): This Secondary Mathematics course is designed to prepare future secondary math educators by focusing on the application of fundamental math concepts, the development of effective teaching strategies, and the alignment of instruction with state content standards. This course teaches key areas including application of number systems, algebraic structures, calculus, geometric reasoning, and probability and statistics in the secondary setting. Additionally, learners will learn to design and evaluate mathematical problems, integrate technology, and differentiate instruction to meet all students’ needs. Learners will also explore how to research and evaluate curricular resources to ensure they align with educational goals and standards. By the end of the course, participants will be equipped with the skills and knowledge necessary to deliver high- quality math instruction in secondary education.